Magnetic Field (Supplementary Problems)

1. A uniform magnetic field, B = 3.0 G, exists in the x-direction. A proton q = +e shoots through the field in the y-direction with a speed of 5.0×10^6 m/s. (a) Find the magnitude and direction of the force on the proton. (b) Repeat with the proton replaced by an electron.

2. The charge shown in Fig. 30-5 is a proton q = + e, $m_p = 1.67 \times 10^{-27}$ kg) with speed 5.0×10^6 m/s. It is passing through a uniform magnetic field directed out of the page; B is 30 G. Describe the path followed by the proton.

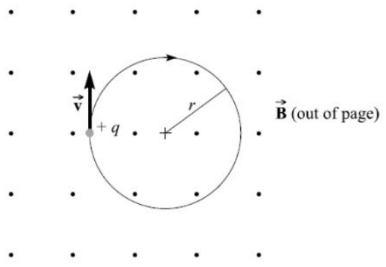


Fig. 30-5

3. A proton enters a magnetic field of flux density 1.5 Wb/m^2 with a velocity of 2.0×10^7 m/s at an angle of 30° with the field. Compute the force on the proton.

4. A cathode ray beam (an electron beam; $m_e = 9.1 \times 10^{-31}$ kg, q = - e) is bent in a circle of radius 2.0 cm by a uniform field with $B = 4.5 \times 10^{-3}$ T. What is the speed of the electrons?

5. As shown in Fig. 30-6, a particle of charge q enters a region where an electric field is uniform and directed downward. Its value E is 80 kV/m. Perpendicular to $\vec{E}$ and directed into the page is a magnetic field B = 0.4 T. If the speed of the particle is properly chosen, the particle will not be deflected by these crossed electric and magnetic fields. What speed should be selected in this case? (This device is called a velocity selector.)

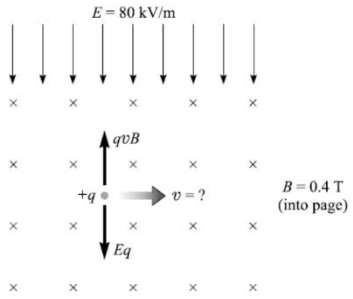
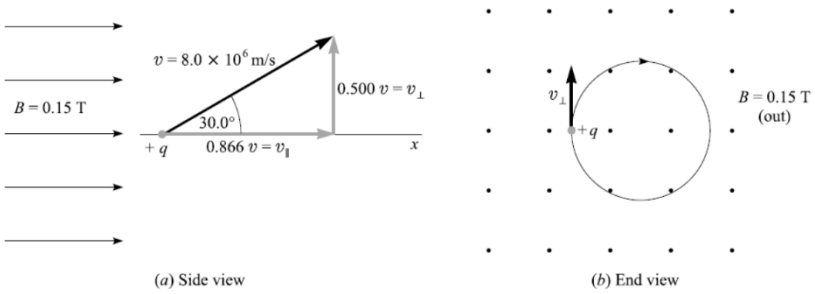


Fig. 30-6

6. In Fig. 30-7 (a), a proton ($q = +e$, $m_p = 1.67 \times 10—27$ kg) is shot with speed 8.0×10^6 m/s at an angle of 30.0° to an x-directed field B = 0.15 T. Describe the path followed by the proton.



(*a*) Side view (*b*) End view

Fig. 30-7

7. Alpha particles $m_\alpha = 6.68 \times 10^{-27}$ kg, q = 2e are accelerated from rest through a p.d. of 1.0 kV. They then enter a magnetic field B = 0.20 T perpendicular to their direction of motion. Calculate the radius of their path.

8. In Fig. 30-8, the magnetic field is out of the page and B = 0.80 T. The wire shown carries a current of 30 A. Find the magnitude and direction of the force on a 5.0 cm length of the wire.

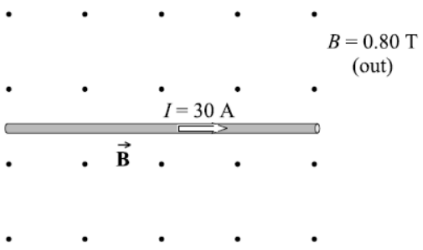


Fig. 30-8

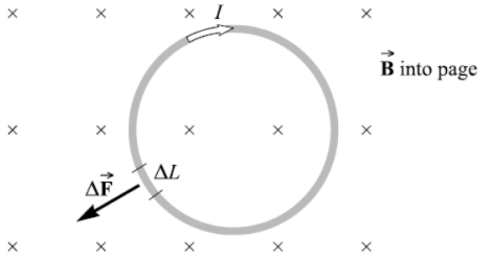


Fig. 30-9

9. As shown in Fig. 30-9, a loop of wire carries a current I and its plane is perpendicular to a uniform magnetic field $\vec{B}$. What are the resultant force and torque on the loop?

10. The 40-loop coil shown in Fig. 30-10 carries a current of 2.0 A in a magnetic field B = 0.25 T. Find the torque on it. How will it rotate?

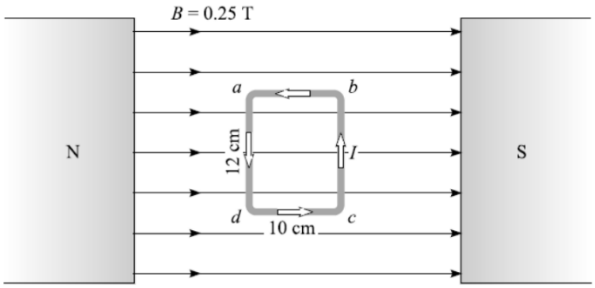
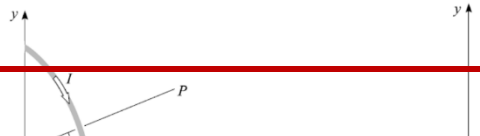


Fig. 30-10

11. In Fig. 30-11 is shown one-quarter of a single circular loop of wire that carries a current of 14 A. Its radius is a 5.0 cm. A uniform magnetic field, B = 300 G, is directed in the x-direction. Find the torque on the loop and the direction in which it



will rotate.

12. Two electrons, both with speed 5.0×10^6 m/s, are shot into a uniform magnetic field $\vec{B}$. The first is shot from the origin out along the + x-axis, and it moves in a circle that intersects the + z-axis at z = 16 cm. The second is shot out along the + y-axis, and it moves in a straight line. Find the magnitude and direction of $\vec{B}$.

13. At a certain place on the planet, the Earth's magnetic field is 5.0×10^{-5} T, directed 40° below the horizontal. Find the force per meter of length on a horizontal wire that carries a current of 30 A northward.

14. An ion (q = +2e) enters a magnetic field of 1.2 Wb/m^2 at a velocity of 2.5 × 105 m/s perpendicular to the field. Determine the force on the ion. Ans. 9.6 × 10—14 N

15. Calculate the speed of ions that pass undeflected through crossed E and B fields for which E = 7.7 kV/m and B = 0.14 T. Ans. 55 km/s

16. The particle shown in Fig. 30-14 is positively charged in all three cases. What is the direction of the force on it due to the magnetic field? Give its magnitude in terms of B, q, and n. Ans. (a) into page, qvB; (b) out of page, qnB sin θ; (c) in plane of page at angle θ + 90°, qvB

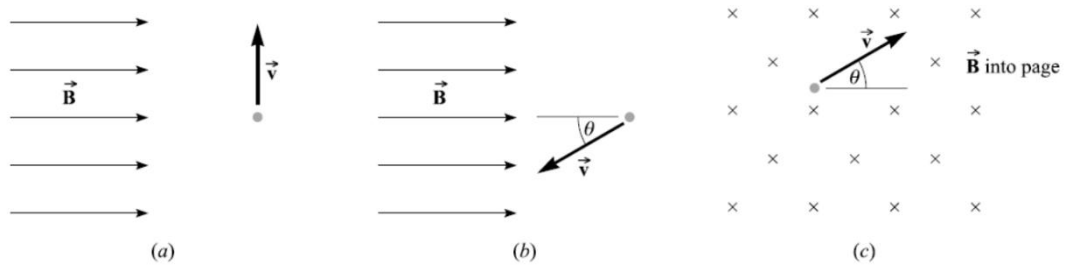


Fig. 30-14

17. What might be the mass of a positive ion that is moving at 1.0×10^7 m/s and is bent into a circular path of radius 1.55 m by a magnetic field of 0.134 Wb/m2? (There are several possible answers.) Ans. n(3.3×10^{-27} kg), where ne is the ion's charge

18. An electron is accelerated from rest through a potential difference of 3750 V. It enters a region where B 4.0 10—3 T perpendicular to its velocity. Calculate the radius of the path it will follow. Ans. 5.2 cm

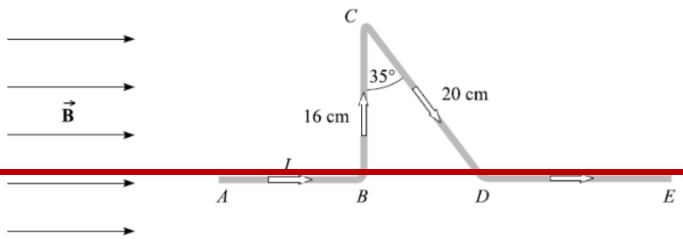
19. An electron is shot with speed 5.0×10^6 m/s out from the origin of coordinates. Its initial velocity makes an angle of 20° to the +x-axis. Describe its motion if a magnetic field B = 2.0 mT exists in the +x-direction. Ans. helix, r = 0.49 cm, pitch = 8.5 cm

20. A beam of electrons passes undeflected through two mutually perpendicular electric and magnetic fields. If the electric field is cut off and the same magnetic field maintained, the electrons move in the magnetic field in a circular path of radius 1.14 cm. Determine the ratio of the electronic charge to the electron mass if E = 8.00 kV/m and the magnetic field has flux density 2.00 mT. Ans. e/me = 175 GC/kg

21. A straight wire 15 cm long, carrying a current of 6.0 A, is in a uniform field of 0.40 T. What is the force on the wire when it is (a) at right angles to the field and (b) at 30° to the field? Ans. (a) 0.36 N; (b) 0.18 N

22. What is the direction of the force, due to the Earth's magnetic field, on a wire carrying current vertically downward? Ans. horizontally toward east

23. Find the force on each segment of the wire shown in Fig. 30-15 if B 0.15 T. Assume the current in the wire to be 5.0 A. Ans. In sections AB and DE, the force is zero; in section BC, 0.12 N into page; in section CD, 0.12 N out of page



24. A rectangular coil of 25 loops is suspended in a field of 0.20 Wb/m^2. The plane of the coil is parallel to the direction of the field. The dimensions of the coil are 15 cm perpendicular to the field lines and 12 cm parallel to them. What is the current in the coil if there is a torque of 5.4 N· m acting on it? Ans. 60 A

25. An electron is accelerated from rest through a potential difference of 800 V. It then moves perpendicularly to a magnetic field of 30 G. Find the radius of its orbit and its orbital frequency. Ans. 3.2 cm, 84 MHz

Magnetic Field (Supplementary Problems)

26. An electron moving along the positive x axis perpendicular to a magnetic field experiences a magnetic deflection in the negative y direction. What is the direction of the magnetic field?

27. A proton travels with a speed of 3.00×10^6 m/s at an angle of 37.0° with the direction of a magnetic field of 0.300 T in the 'y direction. What are (a) the magnitude of the magnetic force on the proton and (b) its acceleration?

28. A proton moves perpendicular to a uniform magnetic field B at 1.00×10^7 m/s and experiences an acceleration of 2.00×10^{13} m/s^2 in the 'x direction when its velocity is in the 'z direction. Determine the magnitude and direction of the field.

29. An electron is accelerated through 2 400 V from rest and then enters a uniform 1.70-T magnetic field. What are (a) the maximum and (b) the minimum values of the magnetic force this charge can experience? A proton moving at 4.00×10^6 m/s through a magnetic field of 1.70 T experiences a magnetic force of magnitude 8.20×10^{13} N. What is the angle between the proton's velocity and the field?

30. A wire 36 m long carries a current of 2.2 A from west to east. Find the magnetic force on the wire if the magnetic field of Earth at this location is directed from south to north and has a magnitude of 5.0×10^{-5} T.

31. A wire 1.0 m long experiences a magnetic force of 0.50 N due to a perpendicular uniform magnetic field. If the wire carries a current of 10.0 A, what is the magnitude of the magnetic field?

32. The magnetic force acting on a wire that is perpendicular to a 1.5 T uniform magnetic field is 4.4 N. If the current in the wire is 5.0 A, what is the length of the wire that is inside the magnetic field?

33. A 2.5 N magnetic force acts on a 475 m wire that is perpendicular to a 0.50 T magnetic field. What is the current in the wire?

34. The magnetic force on a straight 0.15 m segment of wire carrying a current of 4.5 A is 1.0 N. What is the magnitude of the component of the magnetic field that is perpendicular to the wire?

35. An electric power line carries a current of 1400 A in a location where the Earth's magnetic field is 5.0×10^{-5} T. The line makes an angle of 75° with respect to the field. Determine the magnitude of the magnetic force on a 120 m length of wire.

36. Near the equator in South America, the Earth's magnetic field has a strength of 3.2×10^{-5} T. The field at this location is parallel to the surface of the Earth and points due North. A straight wire, 46 m in length, has a east-west orientation and experiences a magnetic force of 0.058 N, directed vertically downward (toward the Earth). What is the magnitude and direction of the current in the wire?

37. t New York City, the Earth's magnetic field has a vertical (downward) component of 5.2×10^{-5} T and a horizontal component of 1.8×10^{-5} T that is directed toward the geographical north. What is the magnitude of the magnetic force on a long, straight wire 6.0 m in length, that carries a 28 A current due East?

38. Compute the value of B in air at a point 5 cm from a long straight wire carrying a current of 15 A.

39. A flat circular coil with 40 loops of wire has a diameter of 32 cm. What current must flow in its wires to produce a field of 3.0×10^{-4} Wb/m^2 at its center?

40. An air-core solenoid with 2000 loops is 60 cm long and has a diameter of 2.0 cm. If a current of 5.0 A is sent through it, what will be the flux density within it?

41. In Bohr's model of the hydrogen atom, the electron travels with speed 2.2×10^6 m/s in a circle $r = 5.3 \times 10^{-11}$ m about the nucleus. Find the value of B at the nucleus due to the electron's motion?

42. A current is carried by a straight conductor in a magnetic field of 1.4 T. The conductor is perpendicular to the magnetic field. The conductor is 2.3 m long, and the magnetic field exerts a force on it.
(a) Calculate the current in the conductor.
(b) What is the angle between the conductor and the magnetic field when the magnetic force is at a maximum? Explain your reasoning.

43. A 120 cm segment of wire has a mass of 0.026 kg. The segment of wire is oriented at a 45° angle to a uniform magnetic field. The strength of the magnetic field is 0.40 T, and the current in the wire segment is 2.3 A.
(a) Calculate the magnitude of the magnetic force on the wire segment.
(b) What is the direction of the magnetic force if the direction of the conventional current is due south and both the wire and the magnetic field are horizontal?

44. A 2.6 m wire carries a current of 2.5 A. The direction of the conventional current is due west. The strength of Earth's magnetic field at the location of the wire is 5.0×10^{-5} T, and the orientation of the field is north.
(a) What are the magnitude and direction of Earth's magnetic force on the wire?
(b) What will the force on the wire be if the wire is rotated to an angle of 72° with the magnetic field?

45. A long, straight wire of length 1.4 m carries a current of I = 3.5 A. A magnetic field of magnitude B = 1.5 T is directed perpendicular to the wire. Calculate the magnitude of the force on the wire.

46. In Figure 9, the current loop forms a right-angled triangle. The loop carries a current I in a uniform magnetic field B directed perpendicular to one edge of the loop. The angle θ in Figure 9 is 39°, L has a length of 1.2 m, and the magnetic field strength has a magnitude of 1.6 T.

(a) What is the force on the part of the loop parallel to the magnetic field?

(b) What is the sum of the forces on the remaining two segments of the loop? (Hint: Write the length of each segment in terms of θ and L, and then use trigonometry.)

(c) What can you conclude about the magnetic force on a closed loop in a uniform magnetic field?

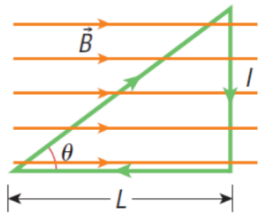


Figure 9

47. A long straight wire coincides with the *x*-axis, and another coincides with the *y*-axis. Each carries a current of 5 A in the positive coordinate direction. (See Fig. 31-3.) Where is their combined field equal to zero?

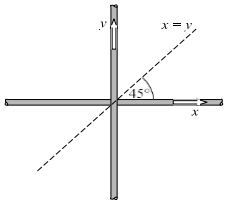


Fig. 31-3

48. A long wire carries a current of 20 A along the axis of a long solenoid. The field due to the solenoid is 4.0 mT. Find the resultant field at a point 3.0 mm from the solenoid axis.

49. As shown in Fig. 31-5, two long parallel wires are 10 cm apart and carry currents of 6.0 A and 4.0 A. Find the force on a 1.0 m length of wire *D* if the currents are (*a*) parallel and (*b*) anti- parallel.

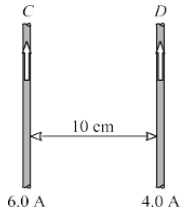


Fig. 31-5

50. Consider the three long, straight, parallel wires shown in Fig. 31-6. Find the force experienced by a 25-cm length of wire *C*.

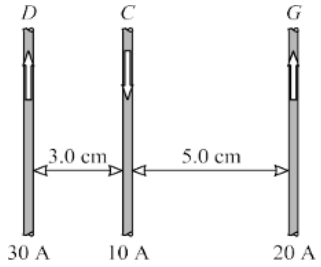


Fig. 31-6

51. A flat circular coil with 10 loops of wire has a diameter of 2.0 cm and carries a current of 0.50 A. It is mounted inside a long solenoid that has 200 loops on its 25-cm length. The current in the solenoid is 2.4 A. Compute the torque required to hold the coil with its axis perpendicular to that of the solenoid.

52. The wire shown in Fig. 31-7 carries a current of 40 A. Find the field at point *P*.

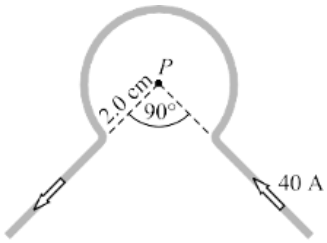


Fig. 31-7

53. Compute the flux density in air at a point 6.0 cm from a long straight wire carrying a current of 9.0 A.
Ans. 30 μT

54. A closely wound, flat, circular coil of 25 turns of wire has a diameter of 10 cm and carries a current of 4.0 A. Determine the value of B at its center. *Ans.* 1.3×10^{-3} Wb/m^2

55. An air-core solenoid 50 cm long has 4000 loops wound on it. Compute B in its interior when a current of 0.25 A exists in the winding. *Ans.* 2.5 Mt

56. A uniformly wound air-core toroid has 750 loops on it. The radius of the circle through the center of its windings is 5 cm. What current in the winding will produce a field of 1.8 mT on this central circle? *Ans*. 0.6 A

57. Two long parallel wires are 4 cm apart and carry currents of 2 A and 6 A in the same direction. Compute the force between the wires per meter of wire length. *Ans.* 6×10^{-5} N/m, attraction

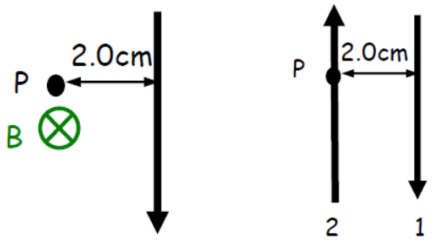
58. Two long fixed parallel wires, *A* and *B*, are 10 cm apart in air and carry 40 A and 20 A respectively, in opposite directions. Determine the resultant field (*a*) on a line midway between the wires and parallel to them and (*b*) on a line 8.0 cm from wire *A* and 18 cm from wire *B*. (*c*) What is the force per meter on a third long wire, midway between *A* and *B* and in their plane, when it carries a current of 5.0 A in the same direction as the current in *A*? *Ans.* (*a*) 2.4×10^{-4} T; (*b*) 7.8×10^{-5} T; (*c*) 1.2×10^{-3} N/m, toward *A*

59. The long straight wires in Fig. 31-3 both carry a current of 12 A, in the directions shown. Find *B* at the point (*a*) x = -5.0 cm, y = 5.0 cm and (*b*) x = -7.0 cm, y = -6.0 cm. *Ans.* (*a*) 96 μT, out; (*b*) 5.7 μT, in

60. A certain electromagnet consists of a solenoid (5.0 cm long with 200 loops) wound on a soft-iron core that intensifies the field 130 times. (We say that the *relative permeability* of the iron is 130.) Find B within the iron when the current in the solenoid is 0.30 A. *Ans.* 0.20 T

61. A certain solenoid (50 cm long with 2000 loops) carries a current of 0.70 A and is in vacuum. An electron is shot at an angle of 10° to the solenoid axis from a point on the axis. (*a*) What must be the speed of the electron if it is to just miss hitting the inside of the 1.6 cm diameter solenoid? (*b*) What is then the pitch of the electron's helical path? *Ans.* (*a*) 1.4×10^7 m/s; (*b*) 14 cm

62. A wire carrying a current of 1.5 A is pointing straight downward as shown on the right. a) Calculate the magnitude of the magnetic field due to the wire at a point, P, 2.0 cm to the left of the wire as shown. b) What is the direction of the magnetic field vector at point P? Show this on the diagram. c) If another wire with a length of 0.75 m and a current of 3.0 A is placed through point, P, pointing straight upward, what is the magnitude of the force it experiences due to wire 1? d) What is the direction of the magnetic force on wire 2? Show this on the diagram. e) Does wire 2 exert a force on wire 1? If so, what is the direction?



63. Two wires are carrying a current of 3.3A are separated by a distance of 6.0 cm. Wire 1 on the left is pointing into the page, and wire 2 on the right is pointing out of the page. A point, P, is 4.0 cm below the center of the two wires. a) Sketch in a circle around each wire with an arrow showing the direction of B. b) Find the magnitude of the magnetic field due to wire 1 at point, P. c) Find the magnitude of the magnetic field due to wire 2 at point, P. d) Find the angle that B_1 is pointing using the geometry of the wires and point. e) Find the x and y components of B_1 and B_2. f) Find the magnitude and direction of the B-field at point, P, due to wire 1 and 2.

Induced EMF: Magnetic Flux (Supplementary Problems)

1. A solenoid is 40 cm long, has cross-sectional area 8.0 cm^2, and is wound with 300 turns of wire that carry a current of 1.2 A. The relative permeability of its iron core is 600. Compute (*a*) *B* for an interior point and (*b*) the flux through the solenoid.

2. The flux through a certain toroid change from 0.65 mWb to 0.91 mWb when the air core is replaced by another material. What are the relative permeability and the permeability of the material?

3. The quarter-circle loop shown in Fig. 32-1 has an area of 15 cm^2. A magnetic field with $B = 0.16$ T exists in the $+x$-direction. Find the flux through the loop in each orientation shown.

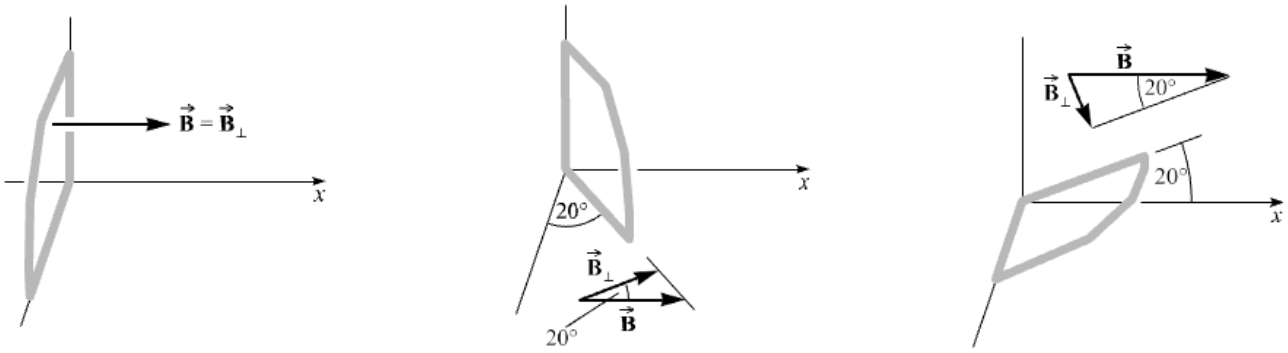
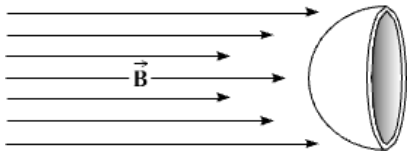


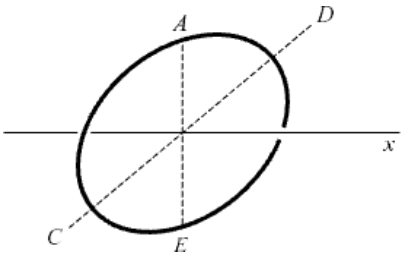
Fig. 32-1

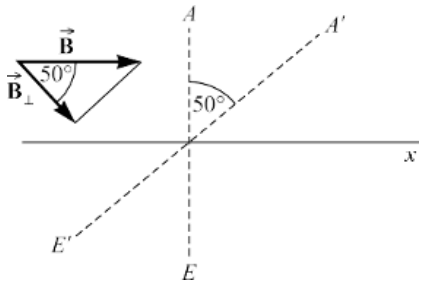
4. A hemispherical surface of radius *R* is placed in a magnetic field B as shown in Fig. 32-2. What is the flux through the hemispherical surface?



5. A 50-loop circular coil has a radius of 3.0 cm. It is oriented so that the field lines of a magnetic field are normal to the area of the coil. Suppose that the magnetic field is varied so that *B* increases from 0.10 T to 0.35 T in a time of 2.0 milliseconds. Find the average induced emf in the coil.

6. In Fig. 32-4(*a*) there is a magnetic field in the2+*x*-direction, with $B = 0.20$ T and a loop of wire in the *yz*-plane. The loop has an area of 5.0 cm and rotates about line *CD* as axis. Point *A* rotates toward positive *x*-values from the position shown. If the loop rotates through 50° from its indicated position, as shown in Fig. 32-4(*b*), in a time of 0.20 s, (*a*) what is the change in flux through the coil, (*b*) what is the average induced emf in it, and (*c*) does the induced current flow from *A* to *C* or *C* to *A* in the upper part of the coil?





7. A coil of 50 loops is pulled in 0.020 s from between the poles of a magnet, where its area intercepts a flux of 3.1×10^{-4} Wb, to a place where the intercepted flux is 0.10×10^{-4} Wb. Determine the average emf induced in the coil.

8. A copper bar 30 cm long is perpendicular to a field of 0.80 Wb/m^2 and moves at right angles to the field with a speed of 0.50 m/s. Determine the emf induced in the bar.

9. As shown in Fig. 32-5, a metal rod makes contact with a partial circuit and completes the circuit. The circuit is perpendicular to a magnetic field with $B = 0.15$ T. If the resistance is 3.0 Ω, how large a force is needed to move the rod as indicated with a constant speed of 2.0 m/s? At what rate is energy dissipated in the resistor?

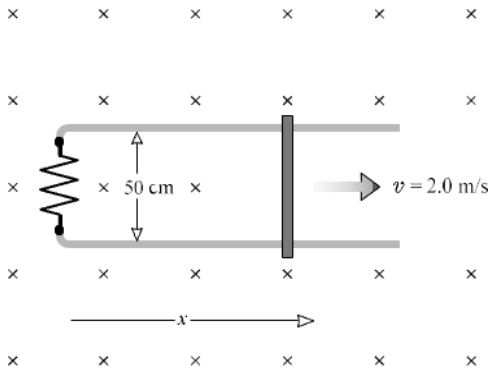


Fig. 32-5

10. The metal bar of length L, mass m, and resistance R shown in Fig. 32-6(*a*) slides without friction on a rectangular circuit composed of resistance less wire on an inclined plane. There is a vertical magnetic field $\vec{B}$. Find the terminal velocity of the bar (that is, the constant velocity it attains).

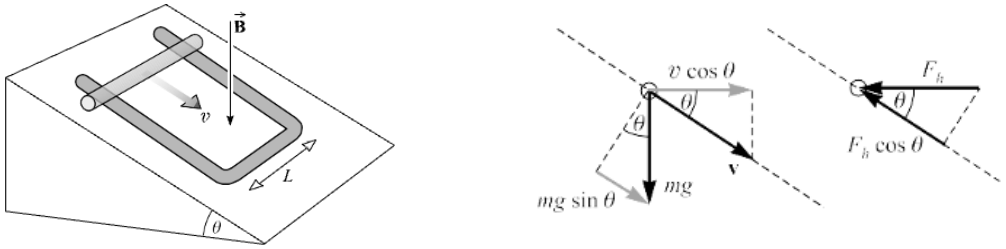


Fig. 32-6

11. The rod shown in Fig. 32-7 rotates about point C as pivot with the constant frequency 5.0 rev/s. Find the potential difference between its two ends, which are 80 cm apart, due to the magnetic field $B = 0.30$ T directed into the page.

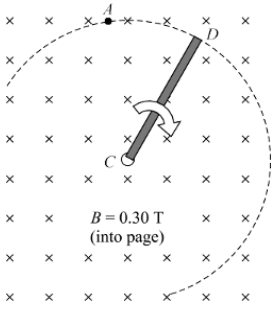


Fig. 32-7

12. A 5.0 Λ coil, of 100 turns and diameter 6.0 cm, is placed between the poles of a magnet so that the flux is maximum through its area. When the coil is suddenly removed from the field of the magnet, a charge of 1.0 10^{-4} C flows through a 595-Λ galvanometer connected to the coil. Compute *B* between the poles of the magnet.

13. A flux of 9.0×10^{-4} Wb is produced in the iron core of a solenoid. When the core is removed, a flux (in air) of 5.0×10^{-7} Wb is produced in the same solenoid by the same current. What is the relative permeability of the iron? *Ans.* 1.8×10^{3}

14. In Fig. 32-8 there is a *x*-directed magnetic field of 0.2 T. Find the magnetic flux through each face of the box shown. *Ans.* Zero through bottom and rear and front sides; through top, 1 mWb; through left side, 2 mWb; through right side, 0.8 mWb.

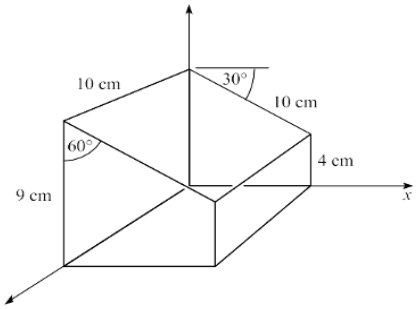


Fig. 32-8

15. A solenoid 60 cm long has 5000 turns on it and is wound on an iron rod of 0.75 cm radius. Find the flux through the solenoid when the current in it is 3.0 A. The relative permeability of the iron is 300. *Ans.* 1.7 m/Wb

16. A room has its walls aligned accurately with respect to north, south, east, and west. The north wall has an area of 15 m^2, the east wall has an area of 12 m^2, and the floor's area is 35 m^2. At the site the Earth's magnetic field has a value of 0.60 G and is directed 50° below the horizontal and 7.0° east of north. Find the fluxes through the north wall, the east wall, and the floor. *Ans.* 0.57 mWb, 56 μWb, 1.6 mWb

17. The flux through the solenoid of Problem 16 is reduced to a value of 1.0 mWb in a time of 0.050 s. Find the induced emf in the solenoid. *Ans.* 67 V

18. A flat coil with radius 8.0 mm has 50 loops of wire. It is placed in a magnetic field $B = 0.30$ T in such a way that the maximum flux goes through it. Later, it is rotated in 0.020 s to a position such that no flux goes through it. Find the average emf induced between the terminals of the coil. *Ans.* 0.15 V

19. The square coil shown in Fig. 32-9 is 20 cm on a side and has 15 loops of wire. It is moving to the right at 3.0 m/s. Find the induced emf (magnitude and direction) in it (*a*) at the instant shown and (*b*) when the entire coil is in the field region. The magnetic field is 0.40 T into the page. *Ans.* (*a*) 3.6 V counterclockwise; (*b*) zero

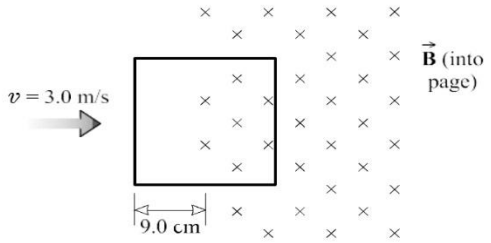
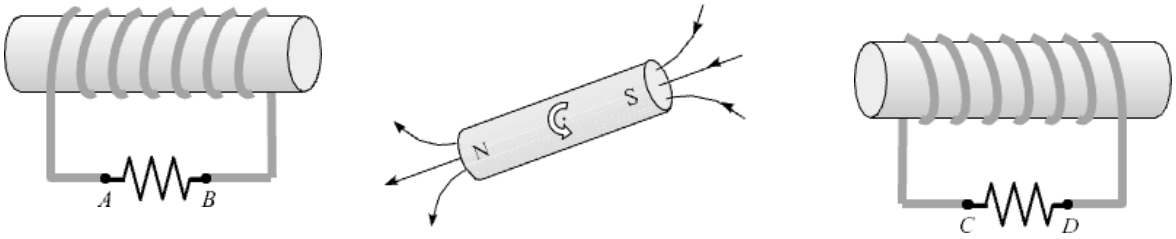


Fig. 32-9

20. The magnet in Fig. 32-10 rotates as shown on a pivot through its center. At the instant shown, in what direction is the induced current flowing (*a*) in resistor *AB*? (*b*) in resistor *CD*? *Ans.* (*a*) *B* to *A*; (*b*) *C* to *D*



21. A train is moving directly south with a speed of 10 m/s. If the downward vertical component of the Earth's magnetic field is 0.54 G, compute the magnitude and direction of the emf induced in a rail car axle 1.2 m long. *Ans.* 0.65 mV from west to east

22. A copper disk of 10 cm radius is rotating at 20 rev/s about its axis and with its plane perpendicular to a uniform field with $B = 0.60$ T. What is the potential difference between the center and rim of the disk? (*Hint*: There is some similarity with Problem 32.12.) *Ans.* 0.38 V

23. How much charge will flow through a 200-Λ galvanometer connected to a 400-Λ circular coil of 1000 turns wound on a wooden stick 2.0 cm in diameter, if a magnetic field B 0.0113 T parallel to the axis of the stick is decreased suddenly to zero? *Ans.* 5.9 μC

24. In Fig. 32-6, described in Problem 32.11, what is the acceleration of the rod when its speed down the incline is n? *Ans.* $g \sin \theta - (B^2L^2v/Rm) \cos^2 \theta$

25. An ac generator produces an output voltage of $\varepsilon = 170 \sin 377t$ volts, where t is in seconds. What is the frequency of the ac voltage?

26. How fast must a 1000-loop coil (each with 20 cm^2 area) turn in the Earth's magnetic field of 0.70 G to generate a voltage that has a maximum value (i.e., an amplitude) of 0.50 V?

27. When turning at 1500 rev/min, a certain generator produces 100.0 V. What must be its angular speed if it is to produce 120.0 V?

28. A certain generator has armature resistance 0.080 Ω and develops an induced emf of 120 V when driven at its rated speed. What is its terminal voltage when 50.0 A is being drawn from it?

29. Some generators, called *shunt generators*, use electromagnets in place of permanent magnets, with the field coils for the electromagnets activated by the induced voltage. The magnet coil is in parallel with the armature coil (it shunts the armature). As shown in Fig. 33-3, a certain shunt generator has armature

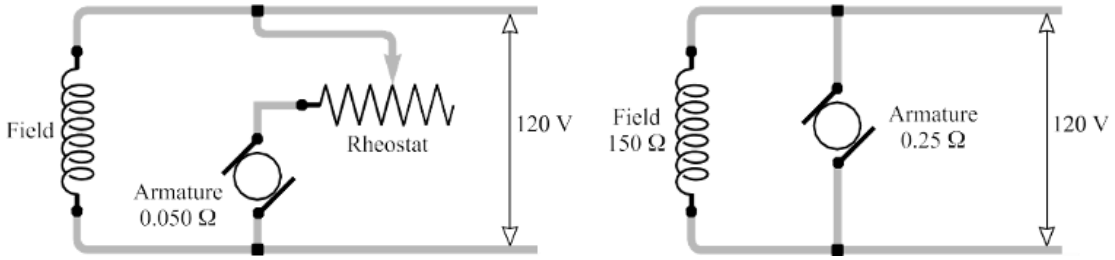


resistance 0.060 Ω and shunt resistance 100 Ω. What power is developed in the armature when it delivers 40 kW at 250 V to an external circuit?

Fig. 33-3

30. The resistance of the armature in the motor shown in Fig. 33-2 is 2.30 Ω. It draws a current of 1.60 A when operating on 120 V. What is its back emf under these circumstances?

31. In a *shunt motor*, the permanent magnet is replaced by an electromagnet activated by a field coil that shunts the armature. The shunt motor shown in Fig. 33-4 has armature resistance 0.050 Λ and is connected to 120 V mains. (*a*) What is the armature current at the starting instant, i.e., before the armature develops any back emf? (*b*) What starting rheostat resistance *R*, in series with the armature, will limit the starting current to б0 A? (*c*) With no starting resistance, what back emf is generated when the armature current is 20 A? (*d*) If this machine were running as a generator, what would be the total induced emf developed by the armature when the armature is delivering 20 A at 120 V to the shunt field and external circuit?



32. A motor has a back emf of 110 V and an armature current of 90 A when running at 1500 rpm. Determine the power and the torque developed within the armature.

33. The emf induced in the armature of a shunt generator is 596 V. The armature resistance is 0.100 Λ. (*a*) Compute the terminal voltage when the armature current is 460 A. (*b*) The field resistance is 110 Ω. Deter- mine the field current, and the current and power delivered to the external circuit.
Ans. (*a*) 550 V; (*b*) 5 A, 455 A, 250 kW

34. A dynamo (generator) delivers 30.0 A at 120 V to an external circuit when operating at 1200 rpm. What torque is required to drive the generator at this speed if the total power losses are 400 W? *Ans.* 31.8 N· m

35. A 75.0-kW, 230-V shunt generator has a generated emf of 243.5 V. If the field current is 12.5 A at rated output, what is the armature resistance? *Ans.* 0.039 9 Ω

36. A 120-V generator is run by a windmill that has blades 2.0 m long. The wind, moving at 12 m/s, is slowed to 7.0 m/s after passing the windmill. The density of air is 1.29 kg/m^3. If the system has no losses, what is the largest current the generator can produce? (*Hint*: How much energy does the wind lose per second?) *Ans.* 77 A

37. A generator has an armature with 500 loops, which cut a flux of 8.00 mWb during each rotation. Compute the back emf it develops when run as a motor at 1500 rpm. *Ans.* 100 V

38. The active length of each armature conductor of a motor is 30 cm, and the conductors are in a field of 0.40 Wb/m^2. A current of 15 A flows in each conductor. Determine the force acting on each conductor. *Ans.* 1.8 N

39. A shunt motor with armature resistance 0.080 Ω is connected to 120 V mains. With 50 A in the armature, what are the back emf and the mechanical power developed within the armature? *Ans.* 0.12 kV, 5.8 kW

40. A shunt motor is connected to a 110-V line. When the armature generates a back emf of 104 V, the armature current is 15 A. Compute the armature resistance. *Ans.* 0.40 Ω

41. A shunt dynamo has an armature resistance of 0.120 Ω. (*a*) If it is connected across 220-V mains and is running as a motor, what is the induced (back) emf when the armature current is 50.0 A? (*b*) If this machine is running as a generator, what is the induced emf when the armature is delivering 50.0 A at 220 V to the shunt field and external circuit? *Ans.* (*a*) 214 V; (*b*) 226 V

42. A shunt motor has a speed of 900 rpm when it is connected to 120-V mains and delivering 12 hp. The total losses are 1048 W. Compute the power input, the line current, and the motor torque. *Ans.* 10 kW, 83 A, 93 N· m

43. A shunt motor has armature resistance 0.20 Ω and field resistance 150 Ω, and draws 30 A when connected to a 120-V supply line. Determine the field current, the armature current, the back emf, the mechanical power developed within the armature, and the electrical eAciency of the machine. *Ans.* 0.80 A, 29 A, 0.11 kV, 3.3 kW, 93%

44. A shunt motor develops 80 N m of torque when the flux density in the air gap is 1.0 Wb/m^2 and the armature current is 15 A. What is the torque when the flux density is 1.3 Wb/m^2 and the armature current is 18 A? *Ans.* 0.13 kN· m

45. A shunt motor has a field resistance of 200 Ω and an armature resistance of 0.50 Λ and is connected to 120-V mains. The motor draws a current of 4.6 A when running at full speed. What current will be drawn by the motor if the speed is reduced to 90 percent of full speed by application of a load?
Ans. 28 A

Inductance; RC and RL Time Constant (Supplementary Problems)

46. A steady current of 2 A in a coil of 400 turns causes a flux of 10^{-4} Wb to link (pass through) the loops of the coil. Compute (*a*) the average back emf induced in the coil if the current is stopped in 0.08 s, (*b*) the inductance of the coil, and (*c*) the energy stored in the coil.

47. A long air-core solenoid has cross-sectional area A and N loops of wire on its length d. (*a*) Find its self-inductance. (*b*) What is its inductance if the core material has permeability μ?

48. A solenoid 30 cm long is made by winding 2000 loops of wire on an iron rod whose cross- sectional area is 1.5 cm^2. If the relative permeability of the iron is 600, what is the self-inductance of the solenoid? What average emf is induced in the solenoid as the current in it is decreased from 0.60 A to 0.10 A in a time of 0.030 s?

49. At a certain instant, a coil with a resistance of 0.40 Λ and a self-inductance of 200 mH carries a current of 0.30 A that is increasing at the rate of 0.50 A/s. (*a*) What is the potential difference across the coil at that instant? (*b*) Repeat if the current is decreasing at 0.50 A/s.

50. A coil of resistance 15 Ω and inductance 0.60 H is connected to a steady 120-V power source. At what rate will the current in the coil rise (*a*) at the instant the coil is connected to the power source, and (*b*) at the instant the current reaches 80 percent of its maximum value?

51. When the current in a certain coil is changing at a rate of 3.0 A/s, it is found that an emf of 7.0 mV is induced in a nearby coil. What is the mutual inductance of the combination?

52. Two coils are wound on the same iron rod so that the flux generated by one passes through the other also. The primary coil has N_p loops and, when a current of 2.0 A flows through it, the flux in it is 2.5×10^{-4} Wb. Determine the mutual inductance of the two coils if the secondary coil has N_s loops.

53. A 2000-loop solenoid is wound uniformly on a long rod with length d and cross-section A. The relative permeability of the iron is k_m. On top of this is wound a 50-loop coil which is used as a secondary. Find the mutual inductance of the system.

54. A certain series circuit consists of a 12-V battery, a switch, a 1.0-MΛ resistor, and a 2.0-μF capacitor, initially uncharged. If the switch is now closed, find (*a*) the initial current in the circuit, (*b*) the time for the

current to drop to 0.37 of its initial value, (*c*) the charge on the capacitor then, and (*d*) the final charge on the capacitor.

55. A 5.0-μF capacitor is charged to a potential difference of 20 kV between plates. After being disconnected from the power source, it is connected across a 7.0-MΛ resistor to discharge. What is the initial discharge current, and how long will it take for the capacitor voltage to decrease to 37 percent of the 20 kV?

56. A coil has an inductance of 1.5 H and a resistance of 0.60 Ω. If the coil is suddenly connected across a 12-V battery, find the time required for the current to rise to 0.63 of its final value. What will be the final current through the coil?

57. A capacitor that has been charged to 2.0×10^5 V is allowed to discharge through a resistor. What will be the voltage across the capacitor after five-time constants have elapsed?

58. A 2.0-μF capacitor is charged through a 30-MΛ resistor by a 45-V battery. Find (*a*) the charge on the capacitor and (*b*) the current through the resistor, both 83 s after the charging process starts.

59. An emf of 8.0 V is induced in a coil when the current in it changes at the rate of 32 A/s. Compute the inductance of the coil. *Ans.* 0.25 H

60. A steady current of 2.5 A creates a flux of 1.4×10^{-4} Wb in a coil of 500 turns. What is the inductance of the coil? *Ans.* 28 mH

61. The mutual inductance between the primary and secondary of a transformer is 0.30 H. Compute the induced emf in the secondary when the primary current changes at the rate of 4.0 A/s.
Ans. 1.2 V

62. A coil of inductance 0.20 H and 1.0-Λ resistance is connected to a 90-V source. At what rate will the current in the coil grow (*a*) at the instant the coil is connected to the source, and (*b*) at the instant the current reaches two-thirds of its maximum value? *Ans.* (*a*) 0.45 kA/s; (*b*) 0.15 kA/s

63. Two neighboring coils, *A* and *B*, have 300 and 600 turns, respectively. A current of 1.5 A in *A* causes 1.2×10^{-4} Wb to pass through *A* and 0.90×10^{-4} Wb to pass through *B*. Determine (*a*) the self-inductance of *A*, (*b*) the mutual inductance of *A* and *B*, and (*c*) the average induced emf in *B* when the current in *A* is interrupted in 0.20 s. *Ans.* (*a*) 24 mH; (*b*) 36 mH; (*c*) 0.27 V

64. A coil of 0.48 H carries a current of 5 A. Compute the energy stored in it. *Ans.* 6 J

65. The iron core of a solenoid has a length of 40 cm and a cross-section of 5.0 cm^2, and is wound with 10 turns of wire per cm of length. Compute the inductance of the solenoid, assuming the relative permeability of the iron to be constant at 500. *Ans.* 0.13 H

66. Show that (*a*) 1 N/A^2 = 1 T· m/A = 1 Wb/A· m = 1 H/m, and (*b*) 1 $C^2/N\cdot m^2$ = 1 F/m.

67. A series circuit consisting of an uncharged 2.0-μF capacitor and a 10-MΩ resistor is connected across a 100-V power source. What are the current in the circuit and the charge on the capacitor (*a*) after one time constant, and (*b*) when the capacitor has acquired 90 percent of its final charge? *Ans.* (*a*) 3.7 μA, 0.13 mC; (*b*) 1.0 μA, 0.18 mC

68. A charged capacitor is connected across a 10-kΩ resistor and allowed to discharge. The potential difference across the capacitor drops to 0.37 of its original value after a time of 7.0 s. What is the capacitance of the capacitor? *Ans.* 0.70 mF

69. When a long iron-core solenoid is connected across a 6-V battery, the current rises to 0.63 of its maximum value after a time of 0.75 s. The experiment is then repeated with the iron core removed. Now the time required to reach 0.63 of the maximum is 0.0025 s. Calculate (*a*) the relative permeability of the iron and (*b*) L for the air-core solenoid if the maximum current is 0.5 A. *Ans.* (*a*) 0.3×10^3; (*b*) 0.03 H

70. What fraction of the initial current still flows in the circuit of Fig. 34-1 seven-time constants after the switch has been closed? *Ans.* 0.000 91

71. By what fraction does the current in Fig. 34-2 differ from i_∞ three time constants after the switch is first closed? *Ans.* $(i_\infty - i)/i_\infty = 0.050$

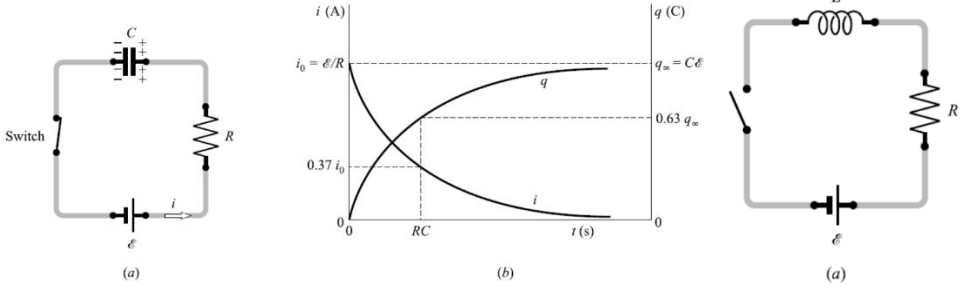


Fig. 34-1

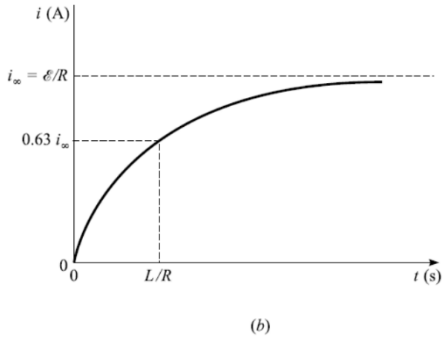


Fig. 34-2

72. In Fig. 34-2, R = 5.0 Ω, L = 0.40 H, and ε = 20 V. Find the current in the circuit 0.20 s after the switch is first closed. *Ans.* 3.7 A

73. The capacitor in Fig. 34-1 is initially uncharged when the switch is closed. Find the current in the circuit and the charge on the capacitor five seconds later. Use R = 7.00 MΩ, C = 0.300 μF, and ε = 12.0 V. *Ans.* 159 nA, 3.27 μC